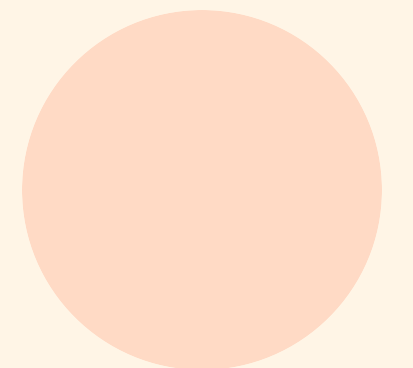


Resonance — Make a Glass Sing!

Let us explore the science of sound!

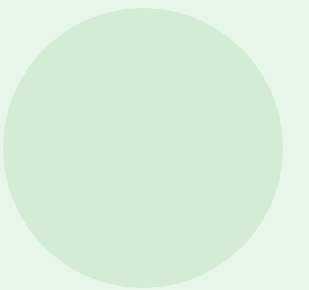
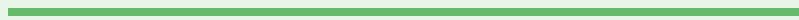


What is Sound?



Sound is a vibration that travels
through air to our ears.

Everything that makes noise is vibrating!

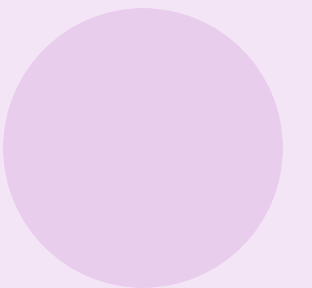
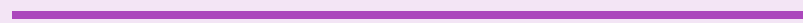


What is Vibration?

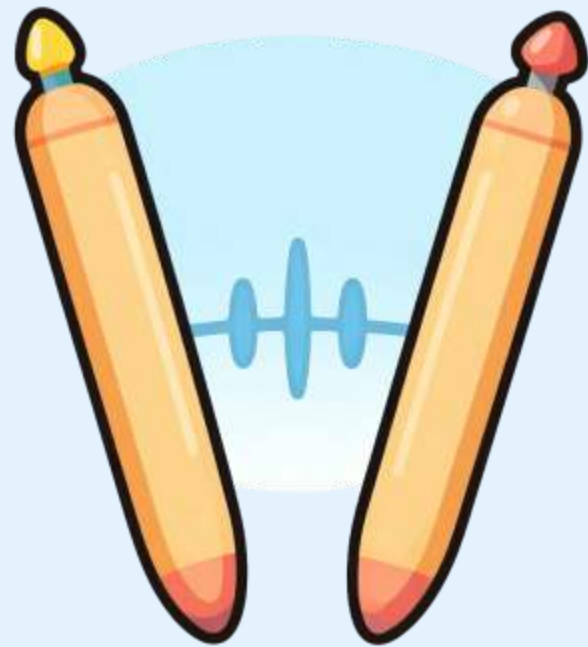
When something moves back and forth very quickly, it vibrates.



Pluck a rubber band — see it shake!
That shaking is vibration.

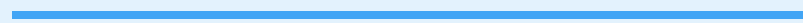


What is Resonance?



Resonance happens when one vibration makes another object vibrate too!

It is like a friend copying your dance moves at the exact same speed.



Real-Life Resonance



A swing goes higher when you push
at just the right time.

That is resonance — matching the
natural rhythm!

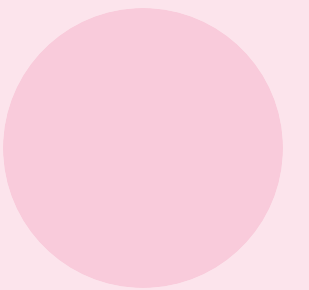


Can a Glass Sing?



Yes! When you rub the rim of a glass with a wet finger, it vibrates.

The vibration makes a musical note.
The glass is singing!

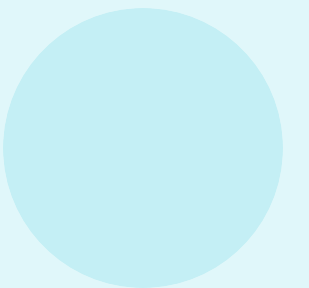
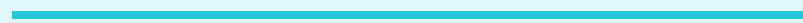


Why Does It Work?



Your wet finger sticks and slips on the glass rim over and over.

This creates a steady vibration that builds up — resonance!

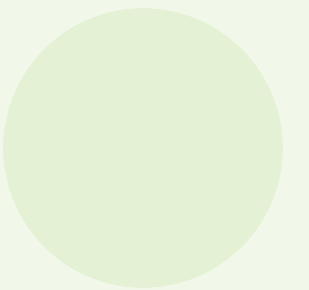


Experiment Time!

Let us make a glass sing!

You will need:

- A thin glass (wine glass shape)
 - Water
 - Clean hands

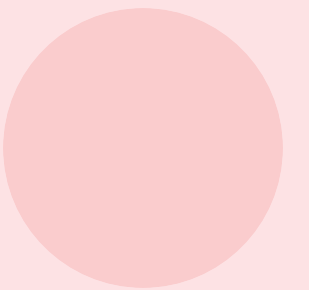


Step 1: Set Up



Fill the glass halfway with water.

Place it on a flat table.
Make sure it does not wobble.



Step 2: Wet Your Finger



Dip your finger in water.

Your finger must stay wet
so it can grip the glass rim.

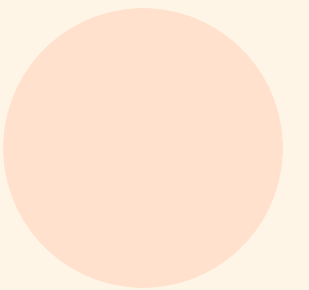


Step 3: Rub the Rim



Gently press your wet finger
on the rim of the glass.

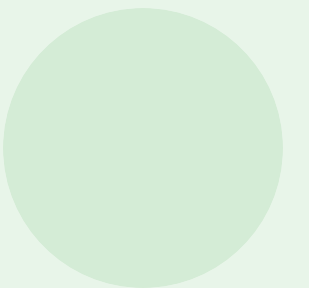
Move it slowly around the rim
in a circle. Keep steady pressure!



Step 4: Listen!

After a few seconds, you will hear
a clear singing note.

The glass is resonating!
Try changing the water level.

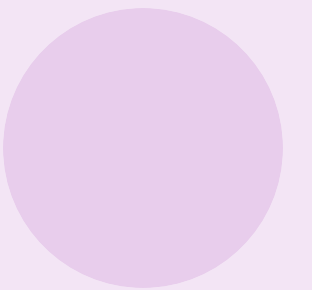
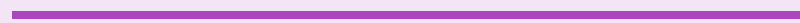


What Changes the Pitch?

More water = lower pitch (deeper sound)
Less water = higher pitch (higher sound)



The water changes how fast
the glass vibrates.

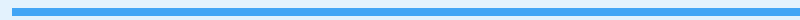


Fun Facts!



Opera singers can break a glass
with their voice alone!

They match the glass's natural
frequency perfectly — resonance!



Safety Tips



Always use a glass with adult help.

Never use a cracked glass.

Clean up any spills right away.

Be gentle — do not press too hard!



Great Job!

You made a glass sing
using resonance!

Now you know that matching vibrations
can create amazing sounds.

